Fuzzy Relations

Crisp relations

\# A crisp relation $R(A, B)$ between two sets A and B is a subset of its cartesian product, $A \times B$.
\# We assign a strength, $\chi(a, b)$, to each ordered pair $(a, b) \in A \times B$ such that:

$$\chi(a, b) = 1 \qquad \text{if } (a, b) \in R$$
$$\chi(a, b) = 0 \qquad \text{if } (a, b) \notin R$$

\# Note that this is a binary relation. For a general n-ary relation $R_n(X_1, X_2, ..., X_n)$, the cartesian product would be all possible n-tuples $(x_1, x_2, ..., x_n)$ such that $x_i \in X_i$.

Fuzzy relations

\# A fuzzy relation $R(A, B)$ of two fuzzy sets A and B with universe of discourse X and Y respectively is an ordered pair $\left((x, y), \mu_{(x,y)}\right)$ such that $(x, y) \in X \times Y$, and $\mu_{(x,y)} \in [0, 1]$.

Fuzzy cartesian product

\# The cartesian product of two fuzzy sets A and B with universe of discourse X and Y respectively, $A \times B$, is:

$$\begin{aligned} A \times B &= \sum_{(x,y) \in X \times Y} \frac{\min(\mu_A(x), \mu_B(y))}{(x, y)} \\ &= \sum_{(x,y) \in X \times Y} \frac{\mu_A(x) \wedge \mu_B(y)}{(x, y)} \end{aligned}$$

Domain and range

\# The *domain* of a fuzzy relation $R(A, B)$ defined on fuzzy sets A and B with universe of discourse X and Y respectively, dom R, is each element in set X with membership values equal to the maximum strength with any element in Y.

$$\text{dom } R = \sum_{x \in X} \frac{\max_{y \in Y} \ \mu_R(x, y)}{x}$$

\# The *range* of a fuzzy relation $R(A, B)$ defined on fuzzy sets A and B with universe of discourse X and Y respectively, ran R, is each element in set Y with membership values equal to the maximum strength with any element in X.

$$\text{ran } R = \sum_{y \in Y} \frac{\max_{x \in X} \ \mu_R(x, y)}{y}$$

\# The *height* of a fuzzy relation $R(A, B)$ defined on fuzzy sets A and B with universe of discourse X and Y respectively, height R, is the maximum strength of any $x \in X$ with any $y \in Y$.

$$h(R) = \max_{x \in X} \ \max_{y \in Y} \ \mu_R(x, y)$$

\# The *inverse* of a fuzzy relation $R(A, B)$ defined on fuzzy sets A and B with universe of discourse X and Y respectively is a $Y \times X$ relation, $R^{-1}(y, x) = R(x, y)$.

Composition

If $R(A, B)$ and $S(B, C)$ are two fuzzy relations defined on fuzzy sets A, B and C with universe of discource X, Y and Z respectively, then composition of R and S defines the relation between A and C, i.e. $R \bigcirc S = T(A, C)$.

Max-min composition

$$\begin{aligned} T(x, z) = R \bigcirc S &= \max_{y \in Y}\ \min(\mu_R(x, y), \mu_S(y, z)) \\ &= \vee_{y \in Y}\ \mu_R(x, y) \wedge \mu_S(y, z) \end{aligned}$$

Max-dot composition

$$\begin{aligned} T(x, z) = R \bigcirc S &= \max_{y \in Y}\ \mu_R(x, y) \cdot \mu_S(y, z) \\ &= \vee_{y \in Y}\ \mu_R(x, y) \cdot \mu_S(y, z) \end{aligned}$$

Example

$$X = \{x_1, x_2\} \qquad Y = \{y_1, y_2\} \qquad Z = \{z_1, z_2, z_3\}$$

$$R = \begin{bmatrix} 0.7 & 0.5 \\ 0.8 & 0.4 \end{bmatrix} \qquad S = \begin{bmatrix} 0.9 & 0.6 & 0.2 \\ 0.1 & 0.7 & 0.5 \end{bmatrix}$$

$$R \bigcirc S = T(x, z) = \begin{bmatrix} 0.7 & 0.6 & 0.5 \\ 0.8 & 0.6 & 0.4 \end{bmatrix}$$

Max-min composition

$$R \bigcirc S = T(x, z) = \begin{bmatrix} 0.63 & 0.42 & 0.25 \\ 0.72 & 0.48 & 0.20 \end{bmatrix}$$

Max-dot composition

Fuzzy equivalence and tolerance

A fuzzy relation $R(A)$, where X is the universe of discourse of fuzzy set A, is equivalent if it is:

Reflexive $\mu_R(x_i, x_i) = 1$

Symmetric $\mu_R(x_i, x_j) = \mu_R(x_j, x_i)$

Transitive $\mu_R(x_i, x_k) \geq \mu_R(x_i, x_j) \wedge \mu_R(x_j, x_k)$

A fuzzy relation $R(A)$ is tolerant if it is reflexive and symmetric, but not necessarily transitive.